

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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1 Introduction

Difference-in-Differences (DiD) has become one of the most popular research designs used to evaluate causal effects of policy interventions. In its canonical format, there are two time periods and two groups: in the first period no one is treated, and in the second period some units are treated (the treated group), and some units are not (the comparison group). If, in the absence of treatment, the average outcomes for treated and comparison groups would have followed parallel paths over time (which is the so-called parallel trends assumption), one can estimate the average treatment effect for the treated subpopulation (ATT) by comparing the average change in outcomes experienced by the treated group to the average change in outcomes experienced by the comparison group. Methodological extensions of DiD methods often focus on this standard two periods, two groups setup; see, e.g., [Heckman et al. \(1997, 1998\)](#), [Abadie \(2005\)](#), [Athey and Imbens \(2006\)](#), [Qin and Zhang \(2008\)](#), [Bonhomme and Sauder \(2011\)](#), [de Chaisemartin and D’Haultfœuille \(2017\)](#), [Botosaru and Gutierrez \(2018\)](#), [Callaway et al. \(2018\)](#), and [Sant’Anna and Zhao \(2020\)](#).¹

Many DiD empirical applications, however, deviate from the canonical DiD setup and have more than two time periods and variation in treatment timing. In this article, we provide a unified framework for average treatment effects in DiD setups with multiple time periods, variation in treatment timing, and when the parallel trends assumption holds potentially only after conditioning on observed covariates. We concentrate our attention on DiD with staggered adoption, i.e., to DiD setups such that once units are treated, they remain treated in the following periods.

The core of our proposal relies on separating the DiD analysis into three separate steps: (i) identification of policy-relevant disaggregated causal parameters; (ii) aggregation of these parameters to form summary measures of the causal effects; and (iii) estimation and inference about these different target parameters. Our approach allows for estimation and inference on interpretable causal parameters allowing for arbitrary treatment effect heterogeneity and dynamic effects, thereby completely avoiding the issues of interpreting results of standard two-way fixed effects (TWFE) regressions as causal effects in DiD setups as pointed out by [Borusyak and Jaravel \(2017\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Sun and Abraham \(2020\)](#), and [Athey and Imbens \(2018\)](#). In addition, it adds transparency and objectivity to the analysis ([Rubin \(2007, 2008\)](#)), and allows researchers to exploit a variety of estimation methods to answer different questions of interest.

The identification step of the analysis provides a blueprint for the other steps. In this paper, we pay particular attention to the disaggregated causal parameter that we call the *group-time average treatment effect*, i.e., the average treatment effect for group g at time t , where a “group” is defined by the time period when units are first treated. In the canonical DiD setup with two periods and two groups, these parameters reduce to the ATT which is typically the parameter of interest in that setup. An attractive feature of the group-time average treatment effect parameters is that they do not directly restrict heterogeneity with respect to observed covariates, the period in which units are first treated, or the evolution of treatment effects over time. As a consequence, these easy-to-interpret causal parameters can be directly used for learning about treatment effect heterogeneity, and/or to construct many other more aggregated causal

¹See Section 6 of [Athey and Imbens \(2006\)](#) and Theorem S1 in [de Chaisemartin and D’Haultfœuille \(2017\)](#) for notable exceptions that cover multiple periods and multiple groups.

parameters. We view this level of generality and flexibility as one of the main advantages of our proposal.

We provide sufficient conditions related to treatment anticipation behavior and conditional parallel trends under which these group-time average treatment effects are nonparametrically point-identified. A unique feature of our framework is that it shows how researchers can flexibly incorporate covariates into the staggered DiD setup with multiple groups and multiple periods. This is particularly important in applications in which differences in observed characteristics create non-parallel outcome dynamics between different groups – in this case, unconditional DiD strategies are generally not appropriate to recover sensible causal parameters of interest (Heckman et al., 1997, 1998; Abadie, 2005). We propose three different types of DiD estimands in staggered treatment adoption setups: one based on outcome regressions (Heckman et al., 1997, 1998), one based on inverse probability weighting (Abadie, 2005), and one based on doubly-robust methods (Sant’Anna and Zhao, 2020). We provide versions of these estimands both for the case with panel data and for the case with repeated cross sections data. To the best of our knowledge, this paper is the first to show how one can allow for covariate-specific trends across groups in DiD setups with variation in treatment timing. Our results also highlight that, in practice, one can rely on different types of parallel trends assumptions and allow some types of treatment anticipation behavior; our proposed estimands explicitly reflect these assumptions.

Our framework acknowledges that in some applications there may be many group-time average treatment effects and researchers may want to aggregate them into different summary causal effect measures. This characterizes the aggregation step of the analysis. We provide ways to aggregate the potentially large number of group-time average treatment effects into a variety of intuitive summary parameters and discuss specific aggregation schemes that can be used to highlight different sources of treatment effect heterogeneity across groups and time periods. In particular, we consider aggregation schemes that deliver a single overall treatment effect parameter with similarities to the ATT in the two period and two group case as well as partial aggregations that highlight heterogeneity along certain dimensions such as (a) how average treatment effects vary with length of exposure to the treatment (event-study-type estimands); (b) how average treatment effects vary across treatment groups; and (c) how cumulative average treatment effects evolve over calendar time. We also provide a formal discussion of the costs and benefits of balancing the sample in “event time” when analyzing dynamic treatment effects. Overall, our setup makes it clear that, in general, the “best” aggregation scheme is application-specific as it depends on the type of question one wants to answer.

Given that our identification results are constructive, we propose easy-to-use plug-in type (parametric) estimators for the causal parameters of interest. Although the outcome regression, inverse probability weighting and doubly-robust estimands are equivalent from the identification point of view, they suggest different types of DiD estimators one can use in practice. Here, we note that using doubly-robust estimators can be particularly attractive as they rely on less stringent modeling conditions than the outcome regression and the inverse probability weighting procedures.

In order to conduct asymptotically valid inference, we justify the use of a computationally convenient multiplier-type bootstrap procedure. This approach can be used to obtain simultaneous confidence bands for the group-time average treatment effects. Unlike commonly used pointwise confidence bands, our simultaneous confidence bands asymptotically cover the *entire path* of the group-time average treatment effects with fixed probability and take into account the dependency across different group-time

average treatment effect estimators. Thus, our proposed confidence bands are arguably more suitable for visualizing the overall estimation uncertainty than more traditional pointwise confidence intervals.

We illustrate the practical relevance of our proposal by analyzing the effect of the minimum wage on teen employment. Here, we follow much empirical work on the effects of the minimum wage and exploit having access to panel data and variation in treatment timing across states (e.g., [Card and Krueger \(1994\)](#); [Neumark and Wascher \(2000, 2008\)](#); [Dube et al. \(2010\)](#), among many others) in order to estimate the effect of the minimum wage on employment. Interestingly, in our setup, using our approach leads to qualitatively different results than results from the TWFE estimator. This suggests that, at least in certain applications, using methods that are robust to treatment effect heterogeneity can lead to meaningful differences relative to more standard TWFE regressions.

Recent Related Literature: This paper is related to the recent and emerging literature on heterogeneous treatment effects in DiD and/or event studies with variation in treatment timing; see, e.g., [de Chaisemartin and D’Haultfoeulle \(2020\)](#), [Goodman-Bacon \(2019\)](#), [Imai et al. \(2018\)](#), [Borusyak and Jaravel \(2017\)](#), [Athey and Imbens \(2018\)](#) and [Sun and Abraham \(2020\)](#). All these papers present, among other things, some negative results about the interpretation of parameters associated with standard TWFE linear regression specifications; see also [Laporte and Windmeijer \(2005\)](#), [Wooldridge \(2005a\)](#), [Chernozhukov et al. \(2013\)](#), and [Gibbons et al. \(2018\)](#) for earlier related results based on (one-way) fixed-effect estimators. Our proposed procedure completely bypasses the pitfalls highlighted in these papers as we clearly separate the identification, aggregation and estimation/inference steps of the analysis.

These aforementioned papers also propose alternative DiD estimators that do not suffer from the pitfalls associated with TWFE. Among these, perhaps the closest to our proposal are those of [de Chaisemartin and D’Haultfoeulle \(2020\)](#), and [Sun and Abraham \(2020\)](#), though several major differences are worth stressing.

[de Chaisemartin and D’Haultfoeulle \(2020\)](#) is focused on recovering an instantaneous treatment effect measure, while we pay particular attention to treatment effect dynamics. In fact, our framework allows one to form *families* of different aggregate parameters in a unified manner. Second, while we pay particular attention to the role played by pre-treatment covariates, [de Chaisemartin and D’Haultfoeulle \(2020\)](#) mainly focus on unconditional DiD designs. On the other hand, the setup in [de Chaisemartin and D’Haultfoeulle \(2020\)](#) is more general than ours as we consider staggered adoption designs and they allow for more general treatment selection. Nonetheless, we note that the unconditional versions of our parallel trends assumptions are weaker than the one in [de Chaisemartin and D’Haultfoeulle \(2020\)](#), even if one were to specialize their setup to staggered adoption designs.

[Sun and Abraham \(2020\)](#) proposes a parameter, cohort-specific average treatment effects, that translates our group-time average treatment effects from calendar time into event time. [Sun and Abraham \(2020\)](#) proposes regression-based estimators of these parameters that have similar properties to our estimators in the specific case of staggered treatment adoption under an unconditional version of the parallel trends assumption. However, our approach is more general in several respects. First, we allow for parallel trends assumptions to hold after conditioning on covariates, and it is not clear how to adapt the regression based estimators in [Sun and Abraham \(2020\)](#) to this case. Second, we consider a wide variety of possible aggregations of group-time average treatment effects where [Sun and Abraham \(2020\)](#) focuses particularly